

DRUG INTERACTIONS—CLINICAL IMPLICATIONS

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Opioids have been shown to have clinically significant interactions with many medications with most interactions occurring through a pharmacokinetic route but some drugs may also interact at a pharmacodynamic level. Pharmacokinetic interactions usually involve the inhibition or induction of hepatic cytochrome P450 (CYP) enzymes. Pharmacodynamic interactions may involve additive (CNS) depression. The role of cytochrome P450 (CYP) enzymes is to oxidise endogenous and exogenous compounds enhancing their water solubility and preparing them for Phase II reactions such as glucuronidation and then elimination from the body. The majority of drugs are metabolised in the liver. There are over 40 CYP enzymes in the human body but only six are responsible for 90% of human drug metabolism: CYP1A2, CYP3A4, CYP2C9, CYP2C19, CYP2D6 and CYP2E1. CYP3A4 is the most important metabolising —50% of drugs. CYP2A6, CYP2B6, CYP3A5 are all thought to play clinically relevant but smaller roles.

Interactions between drugs that are substrates for the same enzymes may result in changes in the rate of metabolism of either or both drugs and may subsequently affect plasma drug concentrations. Drugs metabolised by a common enzyme will compete for metabolic sites and those drugs with a high affinity for an enzyme may inhibit or block the metabolism of other drugs. The CYP enzyme system is greatly affected by such competitive inhibition.

Drugs that inhibit CYP enzyme activity may cause increases in plasma drug concentrations with associated risks of overdose or toxicity. Some drugs are capable of increasing the synthesis of CYP enzymes and therefore increase the number of sites available for biotransformation. This process is termed induction and correspondingly more drug will be metabolised at a time.

Drugs that induce CYP enzymes may increase drug metabolism leading to reduced plasma drug concentrations resulting in reduced efficacy; in the case of opioids it may lead to withdrawal symptoms.

Buprenorphine is metabolised by CYP3A4 but Buprenorphine has a relatively low affinity for CYP3A4 and is therefore a weak inhibitor of the enzyme. To date no significant changes in CYP3A4 activity have been reported in therapeutic concentrations. Although CYP3A4 inhibitors may increase plasma buprenorphine it's safety profile in particular the ceiling effect to respiratory depression means that the effects are unlikely to be life threatening. CYP3A4 inducers may promote rapid buprenorphine metabolism and a subsequent decrease in plasma buprenorphine leading to opiate withdrawal symptoms. While there are studies indicating that buprenorphine also has an inhibitory effect on CYP2D6 in vitro in the body it is not predicted to cause clinically significant interactions with other drugs metabolised by CYP2D6.

Although CYP3A4 was thought to be the primary CYP isoform involved in the metabolism of methadone this is not consistent with human drug interaction studies. Recent data suggest that CYP2B6 plays the most important role in methadone metabolism with CYP2C19 and CYP3A4 being involved to a lesser extent. CYP2C9 and CYP2D6 also have an effect on methadone's metabolism. As so many isoenzymes are involved in the metabolism of methadone methadone interacts with a wide variety of other compounds that use the CYP enzyme system. Co-administration of drugs that inhibit these enzymes may result in reduced methadone metabolism and subsequently elevated serum methadone levels. This could possibly result in problems associated with overdose — in particular potentially fatal respiratory depression. Conversely drugs that induce these CYP enzymes may promote more rapid metabolism of methadone possibly leading to opiate withdrawal symptoms. Close monitoring of the patient may be required.

Pharmacodynamic interactions may occur with concomitant use of buprenorphine or methadone and drugs that suppress respiration such as benzodiazepines. Deaths have been reported when benzodiazepines and buprenorphine

or methadone have been co-abused

Examples of interactions between HIV and opiate therapies will be discussed as they can contribute to non-adherence and poor clinical outcomes. In addition interactions involving benzodiazepines or drugs that may potentiate cardiac effects such as Torsades de Pointes will be considered.

KEY WORDS drug interaction; clinical implication

药物相互作用的临床意义

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摘要 阿片类物质已经被证实能在临床上能与多种药物发生相互作用,多数是药代动力学上的相互作用,也有部分药物是在药效学方面发生的相互作用。药代动力学上的相互作用包括对肝药酶 P450 的抑制或诱导。药效学上的相互作用包括对中枢神经系统附加的抑制作用。抑制肝药酶活性的药物能引起血浆中药物浓度的增高,易导致过量或中毒。诱导肝药酶活性的药物能加速药物的代谢,降低血浆中药物浓度和药物有效性;对阿片物质来说,可能导致戒断症状。药效学的相互作用发生在同时使用抑制呼吸的药物(如苯二氮草类药物)和丁丙诺啡或美沙酮的情况下,两者共同滥用可导致死亡。本文还讨论了 HIV 和阿片治疗之间相互作用的例子,这种相互作用可导致依从性降低以及较差的临床结局。苯二氮草类药物与可加强心血管效应的药物之间的相互作用也需要被考虑到。

关键词 药物相互作用;临床意义

NICOTINE DEPENDENCE AND SMOKING CESSATION

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ABSTRACT Tobacco use is the single most preventable cause of death, disability and disease in the world and is projected to be the leading cause of death and disability across all developed and developing countries by the year 2020. Nicotine, the primary active ingredient of cigarettes that contributes to physical dependence, acts on nicotine receptors in the central nervous system and leads to the release of neurotransmitters (such as dopamine). Like other drugs of abuse, nicotine is thought to, so produces reinforcing effects by activating the mesocorticolimbic dopamine system. A wide variety of cessation treatment of nicotine dependence are commercially available, yet only two general approaches have received empirical validation: behavioral intervention (including 5 As brief intervention) and pharmacotherapy. The evidences show that 5 As brief intervention is one of most cost-effective treatments in clinical work for busy physicians. Three types of medication have been available in market for smoking cessation treatment: nicotine replacement treatment (NRT: ie transdermal patch, gum, inhaler, nasal spray, and lozenge), sustained release bupropion and varenicline. Varenicline, a novel $\alpha 4\beta 2$ nicotinic receptor partial agonist, is efficacious for the treatment of tobacco dependence. The phase 3 trials suggest that it may be more efficacious than NRT and bupropion SR. The safety profile of varenicline is excellent, with the most commonly occurring adverse event, nausea, typically mild and well tolerated. However, new safety warnings were added to the varenicline label because of post-marketing report including agitation, depression and suicidality. A causal connection between varenicline use and these symptoms has not been established.

KEY WORDS tobacco use; nicotine dependence; smoking cessation; behavioral intervention; pharmacotherapy